STUDENT NAME: AMANDEEP	MODULE: My Practice	CODE: SE_002_AGILE
BATCH: ABES_JAS26_FSD JAVA_IIIYear_Batch 02	COURSE: Software Engineering & Agile Framework	STATUS: In Progress

Practice Worksheet – SDLC Models and Agile Sprint Planning

Formulate agile user stories with INVEST criteria, execute Planning Poker story point estimation, facilitate Scrum ceremonies, and perform SDLC comparative evaluation.

KEY LEARNING OBJECTIVES:

- Draft comprehensive Agile User Stories with Given-When-Then Acceptance Criteria.
- Estimate development complexity using Planning Poker and modified Fibonacci scales.
- Facilitate effective Daily Scrums, Sprint Reviews, and Retrospectives.
- Evaluate when to adopt Agile Scrum vs Waterfall vs Kanban methodologies.

STEP-BY-STEP WORKSHEET INSTRUCTIONS:

Step 1: Analyze business requirements and break them down into bite-sized user stories.
Step 2: Validate stories against the INVEST matrix (Independent, Negotiable, Valuable, Estimable, Small, Testable).
Step 3: Calculate sprint capacity based on team velocity and historical commitment ratios.
Step 4: Formulate actionable impediment mitigation strategies in your planning worksheet.

EXERCISE 2.1: AGILE USER STORY DRAFTING WITH INVEST VALIDATION

Problem Statement: An e-commerce portal requires a 'One-Click Reorder' feature for returning customers. Write a complete user story formatted as 'As a... I want... So that...'. Include 3 detailed Acceptance Criteria using the BDD Gherkin syntax (Scenario, Given, When, Then) and validate against each INVEST dimension.

`Reference Baseline / Code / Scenario:`
`As a registered shopper, I want to reorder past purchases in one click so I save checkout time.`

STUDENT SOLUTION & CODE WORKSPACE:
[1] User Story Statement: _____
[2] Given-When-Then Acceptance Criteria: _____
[3] INVEST Criteria Verification: _____

EXERCISE 2.2: PLANNING POKER ESTIMATION & SPRINT CAPACITY ALLOCATION

Problem Statement: A cross-functional team of 5 developers has an average velocity of 35 story points. Given a backlog of 4 incoming epics (OAuth2 Social Login, PDF Invoice Generator, ElasticSearch Catalog, SMS Alerts), assign story points using the modified Fibonacci sequence (1,2,3,5,8,13) and justify complexity vs risk.

Reference Baseline / Code / Scenario:
Feature: OAuth2 Social Login -> Estimate: 5 Story Points (Medium complexity, third-party risk).

STUDENT SOLUTION & CODE WORKSPACE:
[1] Story Point Allocations: _____
[2] Complexity & Risk Justifications: _____
[3] Committed Sprint Backlog: _____

EXERCISE 2.3: SCRUM CEREMONIES & RETROSPECTIVE ACTION PLANNING

Problem Statement: During Sprint 14, the team experienced a 20% commitment spillover due to delayed API dependencies from another team. Design the agenda for a 45-minute Sprint Retrospective using the 'Start, Stop, Continue' framework to address inter-team dependencies and prevent future delivery slippage.

Reference Baseline / Code / Scenario:
Retrospective Framework: What should we START doing? STOP doing? CONTINUE doing?

STUDENT SOLUTION & CODE WORKSPACE:
[1] Ceremony Agenda & Timeboxing: _____
[2] Start / Stop / Continue Action Items: _____
[3] Impediment Escalation Path: _____

EXERCISE 2.4: SDLC COMPARATIVE EVALUATION: SCRUM VS WATERFALL VS KANBAN

Problem Statement: Create a multi-variable decision matrix comparing Scrum, Waterfall, and Kanban across 5 project types: (1) Medical Device Firmware, (2) SaaS Startup MVP, (3) Cloud Infrastructure Maintenance, (4) Banking Core Migration, and (5) E-Commerce Redesign. Recommend the optimal methodology for each with technical rationale.

Reference Baseline / Code / Scenario:
Waterfall is preferred for Medical Device Firmware due to strict FDA regulatory documentation.

STUDENT SOLUTION & CODE WORKSPACE:
[1] Methodology Decision Matrix: _____
[2] Risk & Flexibility Analysis: _____
[3] Recommended SDLC Mappings: _____

SELF-REVIEW & VERIFICATION CHECKLIST:

- [ ] **User Story Depth:** Stories strictly adhere to role, goal, and benefit syntax.
- [ ] **Acceptance Criteria:** Gherkin BDD format provides unambiguous testability.
- [ ] **Estimation Rigor:** Story points calibrated by complexity, uncertainty, and effort.
- [ ] **Ceremony Governance:** Timeboxing and actionable retrospective items documented.
- [ ] **Model Justification:** SDLC selection justified by regulatory and market constraints.

CRITICAL REFLECTION & LEARNING SYNTHESIS:

- *Why is estimating in abstract story points more reliable than estimating in absolute developer hours?*
- *How does a well-crafted 'Definition of Done' (DoD) prevent technical debt from accumulating across sprints?*

Reflection Notes:
_____
_____

COMPLETION STATUS:	EVALUATOR & MENTOR REMARKS:
[ ] COMPLETED (Meets All Solution Criteria) [ ] NEEDS REVISION (Resubmit Corrected Work) **Evaluator Signature:** _____	Feedback Notes: _____ Rubric Evaluation: [ ] Correctness [ ] Code Quality [ ] Best Practices